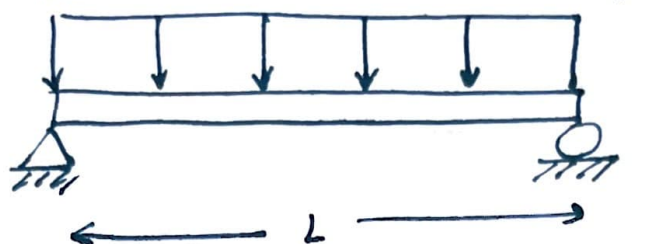


HW4

①

SOLUTION P1:



Find: σ_{xx} & τ_{xy}

Assume: Rectangular cross-section with height h , and width b .

FBD of whole structure:

$$\boxed{\sum F_x = 0} \quad R_{Ax} = 0$$

$$\boxed{\sum F_y = 0} \quad R_{Ay} + R_{By}$$

$$- \int_0^L q_0 dx = 0$$

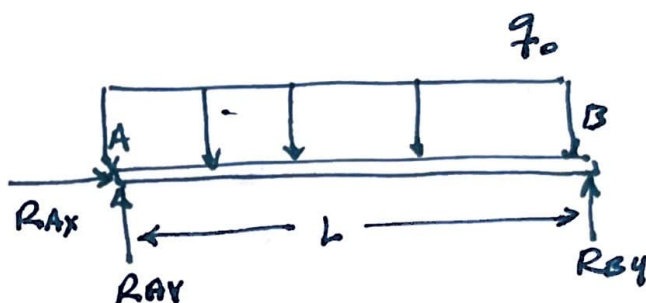
$$\boxed{\sum M_A = 0} \quad L R_{By} - \int_0^L x q_0 dx = 0$$

$$\Rightarrow L R_{By} = q_0 \frac{L^2}{2}$$

$$\Rightarrow \boxed{R_{By} = \frac{q_0 L}{2}}$$

$$\therefore R_{Ay} = q_0 L - q_0 \frac{L}{2}$$

$$\Rightarrow \boxed{R_{Ay} = \frac{q_0 L}{2}}$$



SOLUTION P1 continued:

FBD of part:

$$\boxed{\sum F_x = 0}$$

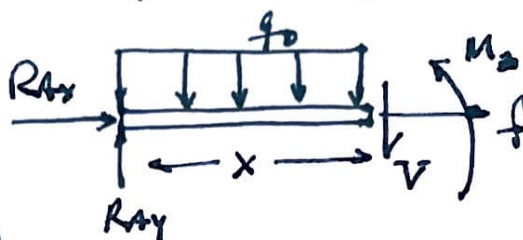
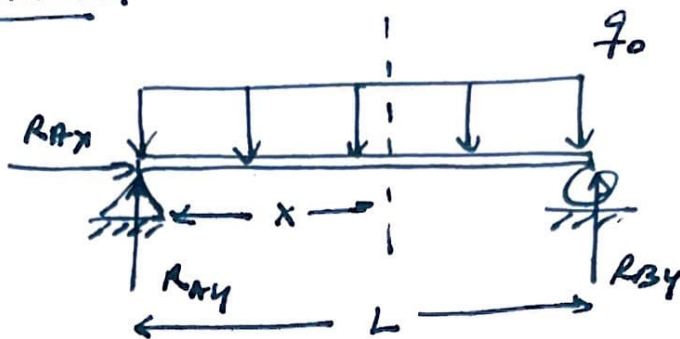
$$R_{Ax} = f$$

$$\Rightarrow f = 0$$

$$\boxed{\sum F_y = 0}$$

$$R_{Ay} - V - \int_0^x q_0 dx = 0$$

$$\Rightarrow V = \frac{q_0 L}{2} - q_0 x$$



$$\boxed{\sum M_A = 0}$$

$$M_z - Vx - \int_0^x q_0 x dx = 0$$

$$\Rightarrow M_z = Vx + \frac{q_0 x^2}{2}$$

$$= \left(\frac{q_0 L}{2} - q_0 x \right) x + \frac{q_0 x^2}{2}$$

$$= \frac{q_0}{2} (Lx - 2x^2 + x^2)$$

$$\boxed{M_z = \frac{q_0}{2} (Lx - x^2)}$$

We know:

$$\sigma_{xx} = - \frac{M_z(x) y}{I_{zz}}$$

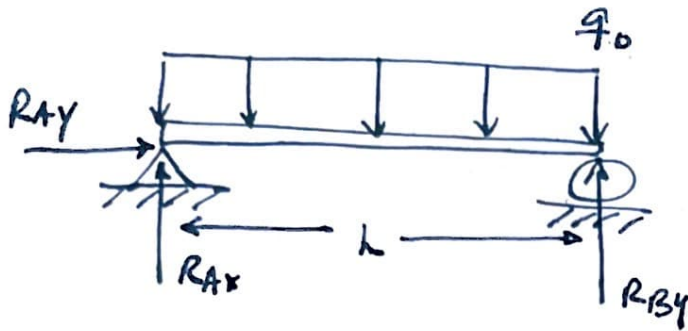
$$\Rightarrow \sigma_{xx} = - \frac{(q_0/2)(Lx - x^2) y}{(bh^3/12)}$$

$$\boxed{\sigma_{xx} = - \frac{6 q_0 (Lx - x^2) y}{bh^3}}$$

$$\Rightarrow \tau_{xy} = \frac{V(x) Q(y)}{I_{zz} b} = \frac{(q_0 L/2 - q_0 x) Q(y)}{(bh^3/12) b}$$

$$\Rightarrow \tau_{xy} = \frac{(3q_0 L - 6q_0 x) \left(\frac{b}{4} \left(\frac{h^2}{4} - y^2 \right) \right)}{bh^3} = \frac{(q_0 L/2 - q_0 x) \left[\frac{b}{2} \left(\frac{h^2}{4} - y^2 \right) \right]}{b^2 h^3}$$

HW4

Solution P2:

Find:

$$\tau'_{xx}|_{\max}$$

$$\tau'_{xy}|_{\max}$$

We know from P1:

$$\tau_{xx}(x,y) = -\frac{6q_0(Lx - x^2)y}{bh^3}$$

$$\tau_{xy}(x,y) = \frac{(3q_0L - 6q_0x)(\frac{h^2}{4} - y^2)}{bh^3}$$

$$\tau_{xx}(x,y) \text{ is max. @ } (\frac{L}{2}, -\frac{h}{2})$$

$$\tau_{xy}(x,y) \text{ is max. @ } (0,0)$$

$$\therefore \tau_{xx}(\frac{L}{2}, -\frac{h}{2}) = -\frac{6q_0(L\frac{L}{2} - \frac{L^2}{4})(-\frac{h}{2})}{bh^3}$$

$$= +\frac{6q_0(\frac{L^2}{4})(\frac{h}{2})}{bh^3} = \frac{\frac{3}{4}q_0L^2h}{bh^3}$$

$$\boxed{\tau_{xx}(\frac{L}{2}, -\frac{h}{2}) = \left(\frac{3}{4}\right) \frac{q_0L^2}{bh^2}}$$

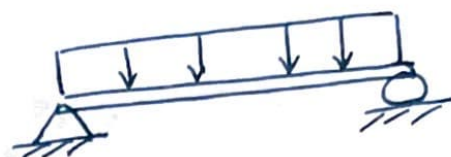
$$\text{Similarly } \tau_{xy}(0,0) = \frac{[3q_0L - 6q_0(0)](\frac{h^2}{4})}{bh^3} = \frac{(\frac{3}{4})(q_0L)}{bh}$$

$$\boxed{\tau_{xy}(0,0) = \frac{3q_0L}{bh}}$$

Solution P3:

$$EI_{zz} \frac{d^4 v}{dx^4} = q(x) \quad \leftarrow \text{Bending equation}$$

Given:



$$q(x) = -q_0$$

$$\Rightarrow \frac{d^4 v}{dx^4} = \frac{-q_0}{EI_{zz}} \quad \rightarrow (2)$$

$\left. \begin{array}{l} \text{4th order ODE} \\ \therefore \text{Need 4 BCs} \end{array} \right\}$

Integrating Equation (2):

$$\frac{d^4 v}{dx^4} = \frac{-q_0}{EI_{zz}}$$

$$\frac{d^3 v}{dx^3} = \left(\frac{-q_0}{EI_{zz}} \right) (x + C_1)$$

$$\frac{d^2 v}{dx^2} = \left(\frac{-q_0}{EI_{zz}} \right) \left(\frac{x^2}{2} + C_1 x + C_2 \right) = \frac{M(x)}{EI_{zz}} \quad \rightarrow (3)$$

$$\frac{dv}{dx} = \left(\frac{-q_0}{EI_{zz}} \right) \left(\frac{x^3}{6} + C_1 \frac{x^2}{2} + C_2 x + C_3 \right)$$

$$v(x) = \left(\frac{-q_0}{EI_{zz}} \right) \left(\frac{x^4}{24} + C_1 \frac{x^3}{6} + C_2 \frac{x^2}{2} + C_3 x + C_4 \right) \quad \rightarrow (4)$$

Boundary conditions:

(i) $v(x=0) = 0$ at $x=0$
 (iii) $v(L) = 0$ at $x=L$

(ii) $\frac{d^2 v}{dx^2}(x=0) = 0$ at $x=0$

(iv) $\frac{d^2 v}{dx^2}(x=L) = 0$ at $x=L$

BC (i): $v(0) = 0$ at $x = 0$

$$\Rightarrow v(0) = \frac{-q_0}{EI_{zz}} \left(\frac{x^4}{24} + C_1 \frac{x^3}{6} + C_2 \frac{x^2}{2} + C_3 x + C_4 \right) = 0$$

$\Rightarrow C_4 = 0 \rightarrow (5)$

BC (ii): $\frac{d^2 v}{dx^2}(x=0) = 0$ at $x = 0$

$\Rightarrow$ From equation (3) $\frac{d^2 v}{dx^2}(0) = \left(\frac{-q_0}{EI_{zz}} \right) \left(\frac{x^2}{2} + C_1 x + C_2 \right) = 0$

$\Rightarrow C_2 = 0 \rightarrow (6)$

BC (iii): $v(L) = 0$ at $x = L$

$$\Rightarrow v(L) = \left(\frac{-q_0}{EI_{zz}} \right) \left(\frac{L^4}{24} + C_1 \frac{L^3}{6} + C_2 \frac{L^2}{2} + C_3 L + C_4 \right) = 0$$

$$\Rightarrow \frac{L^4}{24} + C_1 \frac{L^3}{6} + C_3 L = 0$$

$$\Rightarrow \frac{L^3}{24} + C_1 \frac{L^2}{6} + C_3 = 0$$

$$C_3 = -\frac{L^3}{24} - C_1 \frac{L^2}{6} \rightarrow (7)$$

BC (iv): $\frac{d^2 v}{dx^2}(x=L) = 0$ $\Rightarrow$ From eq (3) $\frac{d^2 v}{dx^2}(L) = \left(\frac{-q_0}{EI_{zz}} \right) \left(\frac{L^2}{2} + C_1 L + C_2 \right) = 0$

$$\Rightarrow \left(\frac{-q_0}{EI_{zz}} \right) \left(\frac{L^2}{2} + C_1 L + C_2 \right) = 0 \Rightarrow C_1 = -\frac{L}{2} \rightarrow (8)$$

$\Rightarrow$ From eq (7) $C_3 = -\frac{L^3}{24} - \left(-\frac{L}{2} \right) \left(\frac{L^2}{6} \right) \Rightarrow C_3 = \frac{L^3}{12} - \frac{L^3}{24}$

$$\Rightarrow C_3 = \frac{L^3}{24} \rightarrow (9)$$

Finally we have an expression for $v(x)$:

$$v(x) = \left(\frac{-q_0}{EI_{zz}} \right) \left[\frac{x^4}{24} + C_1 \frac{x^3}{6} + C_2 \frac{x^2}{2} + C_3 x + C_4 \right]$$

$$= \left(\frac{-q_0}{EI_{zz}} \right) \left[\frac{x^4}{24} + \left(-\frac{L}{2} \right) \frac{x^3}{6} + \left(\frac{L^3}{24} x \right) \right]$$

$$\Rightarrow v(x) = \left(\frac{-q_0}{24 EI_{zz}} \right) (x^4 - 2Lx^3 + L^3 x) \rightarrow (10)$$

(6)

Verify: $v(0) = 0$ using eq (10)

$$v(0) = \left(\frac{-q_0}{24EI_{zz}} \right) \left[0 - \cancel{2L(0)} + L(0) \right]$$

$$\Rightarrow v(0) = 0 \checkmark$$

Verify: $v(L) = 0$ using eq (10)

$$v(L) = \left(\frac{-q_0}{24EI_{zz}} \right) \left[L^4 - 2L^3 + L^3 \right]$$

$$= \left(\frac{-q_0}{24EI_{zz}} \right) \left[L^4 - \cancel{2L^4} + L^4 \right] \Rightarrow v(L) = 0 \checkmark$$

Finally on equation (10): $v(x) = \left(\frac{-q_0}{24EI_{zz}} \right) [x^4 - 2Lx^3 + L^3x]$

@ $x = \frac{L}{2}$ we have

$$v\left(\frac{L}{2}\right) = \left(\frac{-q_0}{24EI_{zz}} \right) \left[\left(\frac{L}{2}\right)^4 - 2L\left(\frac{L}{2}\right)^3 + L^3\left(\frac{L}{2}\right) \right]$$

$$= \left(\frac{-q_0}{24EI_{zz}} \right) \left[\frac{L^4}{16} - \frac{2L^4}{8} + \frac{L^4}{2} \right] < 0$$

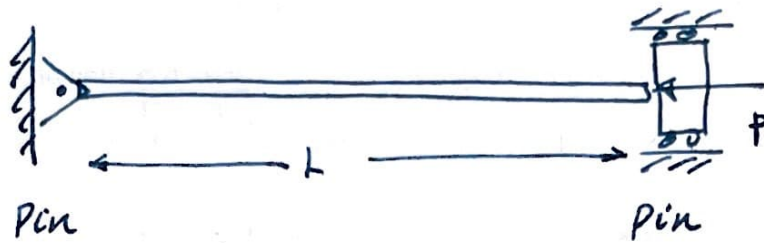
We can say that $0 \leq x \leq L, v(x) < 0$

$\therefore$ The structure bends in the downward direction.

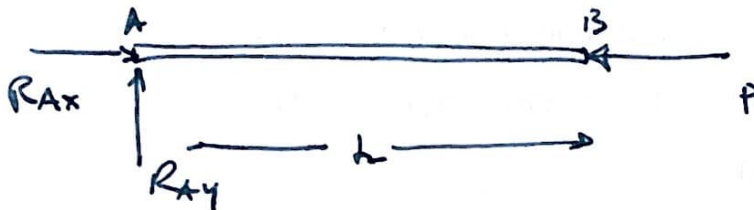


(4)

HW 4

SOLUTION P4:

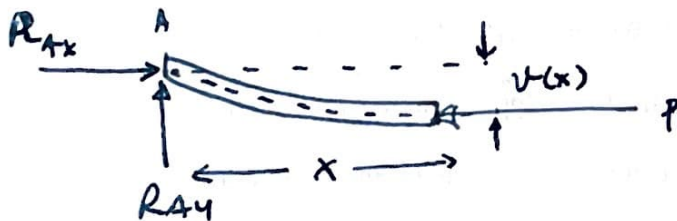
pin-ended column
or
pinned-pinned column

FBD of the whole system:

$$\boxed{\sum F_x = 0} \quad R_{Ax} = P$$

$$\boxed{\sum F_y = 0} \quad R_{Ay} = 0$$

$$\boxed{\sum M_A = 0}$$

FBD of part: (under Buckling)

$$\boxed{\sum F_x = 0} \quad R_{Ax} = P$$

$$\boxed{\sum F_y = 0} \quad R_{Ay} = 0$$

$$\boxed{\sum M_A = 0} \quad M_z = -Pv(x)$$

Deflection equation:

$$EI_{zz} \frac{d^2 v}{dx^2} = -P[v(x)]$$

$$\Rightarrow \frac{d^2 v}{dx^2} = \left(\frac{-1}{EI_{zz}} \right) (Pv) \Rightarrow \boxed{\frac{d^2 v}{dx^2} + \frac{Pv}{EI_{zz}} = 0}$$

$$\left\{ \text{where } k^2 = \frac{P}{EI_{zz}} \right\}$$

②

We need to find solutions for

$$\boxed{\frac{d^2 v}{dx^2} + k^2 v = 0}$$

$$\boxed{v(x) = C_1 \cos kx + C_2 \sin kx} \rightarrow \text{general solution}$$

Boundary conditions : i) $\boxed{v(0) = 0}$

$$\Rightarrow 0 = (C_1) \cos(0) + (C_2) \sin(0) \Rightarrow \boxed{C_1 = 0}$$

$$(ii) \boxed{v(L) = 0}$$

$$\Rightarrow 0 = C_2 \sin kL \Rightarrow \sin kL = 0$$

$$\text{or } kL = n\pi$$

$$\Rightarrow k = \frac{n\pi}{L} \Rightarrow k^2 = \frac{n^2\pi^2}{L^2}$$

$$\therefore \text{max def } \frac{P}{EI_{zz}} = \frac{n^2\pi^2}{L^2}$$

$$\boxed{\text{For } P = P_{cr} ; n = 1}$$

$$\Rightarrow \boxed{P_{cr} = \frac{EI_{zz}\pi^2}{L^2}}$$